SIMATS ENGINEERING

Name	Register Number	Course Code	Course
S Susmitha	192571068	ITA0510	Computer Vision

35. Morphological operations based on OpenCV using Black hat technique.

AIM:
Morphological operations using Black hat technique.

PROGRAM:

```
import cv2

# Create a 3x3 rectangular structuring element
filterSize = (3, 3)
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, filterSize)

# Read the image in grayscale
input_image = cv2.imread(
    r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png",
    cv2.IMREAD_GRAYSCALE
)

if input_image is None:
    print("Error: Image not found!")
else:
    # Apply Black Hat transformation
    blackhat_img = cv2.morphologyEx(input_image, cv2.MORPH_BLACKHAT, kernel)

    # Save the output image
    cv2.imwrite(
        r"C:\Users\Susmitha\Downloads\ITA0510 Lab\blackhat.jpg",
        blackhat_img
    )

    # Display the images
    cv2.imshow("Original Image", input_image)
    cv2.imshow("Black Hat Image", blackhat_img)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

INPUT:

OUTPUT:

RESULT:

Thus the program is executed successfully.